

Aim:- Use of Basic Unix Shell commands: ls, mkdir, rmdir, cd, cat, banner, touch, file, wc, sort, cut, grep, dd, dfspace, du, ulimit.

### Theory

#### Introduction to Linux

LSP → linux shell programming

OS → Operating Software is a system software which provide a interface between user & hardware computer.

Basically software has three type - Application Software, utility software & System software.

Exemples of System operating software -

windows, andriod, linux, ios, MAC, unix etc.

\* Unix It is a ~~first~~ operating system that is powerfull & portable.

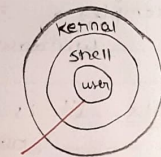
→ It is multiuser, multitasking & interface provide between user & computer hardware.

→ Developed in 1969-70. by Ken Thompson & Dennis mitchie (C programming founder) at AT & T bell lab.

→ It is a license based operating system controlled by company with private access.

→ It is not an open source





Architecture

\* Linux linux is an open-source operating system based on Unix.

→ It was developed by Linus Torvalds in 1991.

→ Widely used in servers, personal computers, and embedded systems because of its stability, security & multitasking capability.

→ Multitask & Multitasking, interface ~~from~~ between user & hardware.

→ Uses ⇒ Server banking, bit data, Railways, defence, Refs, Research lab, android etc.

→ Linux follows a hierarchical file system structure, where all files & directories are organised starting from the root directory (/).

### Architecture of Linux

Shell :- Translator

Shell is a command line interpreter. It provides interface between user & kernel.

→ It work at user level.

| Unix                         | < Linux                      | < Windows                                                   | [based on user comfort] |
|------------------------------|------------------------------|-------------------------------------------------------------|-------------------------|
|                              | command line interface (CLI) | Graphical user interface (GUI)                              |                         |
| It is fully based on command |                              | It's partially based on mouse click control (user friendly) |                         |

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Kernal :- Kernal is a heart of operating system.

- Kernal is a core part of OS that is responsible for managing system resources.
- Interface provider between shell & hardware.
- It works at system level.

Used in process management, memory management & device control.

→ File system :- Organizes files & directories.

→ Utilities & Command :- Programs used to perform various tasks.

→ The Shell accepts commands from user & executes them. The most commonly used shell in Linux is the Bash (Bourne Again shell)

→ Android mobile uses linux - Kernal part to process & memory management.

Difference between Linux & Windows

| Windows                        | Linux                      |
|--------------------------------|----------------------------|
| Desktop / Folder / Recycle bin | Screen / Directory / trash |

|                                   |                          |
|-----------------------------------|--------------------------|
| Command terminal ⇒ Ctrl + Alt + T | POST → Pre on self start |
|-----------------------------------|--------------------------|

student @ ubuntu : ~ \$

↓  
username

↓  
System

↓  
Home directory

↓  
create command

firmware → BIOS

(Basic input/output system)

multiple OS works simultaneously  
virtualization.

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Basic Unix commands help users perform operations-

1. `ls` → Displays the list of files & directories.  
Syntax → `ls [options] [directory]`

Ex

`ls`

`ls -l`

`ls -a`

2. `mkdir` → Creates new directory.  
Syntax → `mkdir name`

3. `rmdir` → Removes an empty directory.  
Syntax → `rmdir name`

4. `cd` → changes the current directory  
`cd directory name`  
`cd ..` ⇒ back to previous directory.

5. `cat` → Displays the contents of a file  
Syntax → `cat fileName.ext`

6. `banner` → Prints characters in large ASCII format.  
Ex `banner HELLO`  
`banner HELLO`

7. `touch` → Creates an empty file  
Syntax `touch fileName`  
Ex `touch file.txt`

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## Sample Commands

Kench> cd documents

Kencl Documents> cd linux

Kencl Documents\Linux> ls

file1.txt file2.txt temp temp.txt

Kencl Documents\Linux> ls -l

total 4

-rw-r--r-- 1 Kencl 197603 48 May 17 10:30 file1.txt

-rw-r--r-- 1 Kencl 197603 18 May 17 10:12 file2.txt

-rw-r--r-- 1 Kencl 197603 265 May 17 10:45 temp

-rw-r--r-- 1 Kencl 197603 265 May 17 10:46 temp.txt

Kencl Documents\Linux> mkdir testdir

Kencl Documents\Linux> ls

file1.txt file2.txt temp temp.txt testdir

Kencl Documents\Linux> cd testdir

Kencl Documents\Linux\testdir> pwd

Kencl Documents\Linux\testdir

Kencl Documents\Linux\testdir> touch file3.txt

Kencl Documents\Linux\testdir> file file3.txt

file3.txt: ASCII text, with CRLF line terminators

Kencl Documents\Linux\testdir> cd ..

Kencl Documents\Linux>

8. file → Displays the type of a file.

Syntax file filename.txt

output → text

9. wc → Counts the number of lines, words & characters

Syntax → wc filename

→ m y z  
↓ ↓ ↓  
word word character  
line

wc -l file.txt ⇒ number of lines

wc -w file.txt ⇒ number of words

10. Sort → Sorts lines in alphabetical order.

Syntax sort filename.

11. cut → extracts specific columns from a file

Syntax cut -d 'delimiter' -f field filename.

12. grep → Searches for patterns in files

Syntax grep "hello" file.txt

13. dd → Copies or converts files

Syntax dd if = file1.txt of = file2.txt  
↓ ↓  
input file output file

14. df → displays disk space usage

Syntax df  
df -h



15. clear → for clear screen

16. cal → for calendar

17. date → for date & time

18. who → Display information about users who currently logged into the system.

Syntax ⇒ who

who -q ⇒ number of users.

who -H ⇒ for showing headline

19. mv → moves file or directories from one location to another. Also used to rename files.

Syntax ⇒ mv source destination.

Ex

mv oldname.txt /home/user/Documents (move)

mv oldname.txt newname.txt (rename)

mv folder1 folder2 (move into folder2)

20. cp ⇒ copy files or directories from one location to another.

Syntax cp source destination

Q cp file1.txt file2.txt

cp -r folder1 folder2 (copy not move)

Result :- Thus, the basic Unix/Linux shell commands were successfully executed & their outputs were observed.



## Viva Questions -

Q1 What is Linux?

Ans Linux is an open source, Unix-like operating system, developed by Linus Torvalds in 1991. It is widely used for servers, computers & embedded systems because of its stability & security with multitasking.

Q2 What is a shell in Linux?

Ans A shell is a command-line interpreter that allows users to interact with the Linux operating system by executing commands & scripts.

Q3 What does the ls command do?

Ans The ls command lists files & directories in the current directory. Options like `-l` show detailed information & `-a` show hidden files.

Q4 What is the use of the grep command?

Ans The grep command searches for a specific pattern or word in a file & displays the matching lines.

Q5 What is the difference between cp & mv commands?

Ans 'cp' command copies files or directories to another location, keeping the original. 'mv' moves or renames files & removes them from the original location.



Aim To study command related to inode, I/O redirection (& piping, process control commands, mails.

Theory - Linux provides powerful command-line tools to control files, process & communication between program.

Redirection can be defined as changing the way from where commands read input to where commands send output. We can redirect input and output command.

For redirection, meta characters are used. It can be into a file (by '<', '>') or a program (by pipe symbol '|')

### Standard Stream I/O Redirection

The bash shell has three standard stream I/O redirection:-

1) Standard input (stdin) - numbered as stdin(0)  
bash takes input from stdin.

By default ~~keyboard~~ keyboard is used as input.

2) Standard output (stdout) - numbered as stdout(1)  
bash shell sends output to stdout.  
By default output goes to screen/display.

3) Standard error (stderr) - numbered as stderr(2)  
Bash shell sends error message to stderr.  
error goes to display.

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\* Inode An inode (index node) is a data structure in Linux that stores information about a file except its name. Each file in Linux has a unique inode number.

Information stored in inode includes -

- File type
- File size
- Owner information
- Permission
- Time of creation or modification
- Location of file data on disk

The `ls-i` command is used to display the inode number of a file.

Ex `ls -i file1.txt`

\* File redirection :-

- > => Redirect output to file
- >> => Append output to file (did not overwrite)
- < => Take input from file
- | => pipe output to another command
- << => read from standard input until "String" is encountered as the only thing on a line
- 2> => error

Ex `echo "Hello Linux" > file1.txt`

`tail -n +4` => Jump to 4 lines & print

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\* Piping :- A pipe (|) is used to connect the output of one command as the input to another command. It allows multiple commands to work together.

Syntax

Command 1 | Command 2

Ex

ls | sort

This command sends the output of ls to the sort command as input.

Note

stat file.txt

it gives file name  
modify  
Birth  
Size  
f.

} meta data of file.

ls > file.txt

echo "Hello" > file.txt

echo "Hi!" > file.txt ~~will~~ overwrite the existing input data from

ls echo "Hi!" >> file.txt -> add input in next line (append)

cat < file.txt => ~~is~~ read the content of file)



\* Process Control Commands:- A process is a program that is currently running in the system. Linux provides commands to manage processes.

Common process control commands:-

ps → Display currently running process.

top → shows running processes in real time.

kill → Terminates a running process using its process id. (PID)

Ex kill 1234

bg → Runs a ~~suspended~~ process in the background.

fg → Brings a ~~background~~ process to the foreground.

\* Mail Command:-

Linux provides the mail command to send and receive messages between user with the help of user ID.

Syntax

mail username

Ex mail user1

After typing the message, Ctrl + D to send the message.

Result :- Thus, Linux commands, related to inode, I/O redirection, piping, process control, & mail were successfully studied & executed.

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Viva Question

Q1 What is an inode in Linux?

Ans An inode is a data structure that stores information about a file such as permission, owner, size & location on disk, except the file name.

Q2 What is I/O redirection?

Ans I/O redirection is used to change the standard input or output of a command using operators like  $>$ ,  $>>$ ,  $<$  &  $<<$ .

Q3 What is a pipe in Linux?

Ans A pipe ( $|$ ) in Linux, connects the output of one command to the input of another command, allowing multiple commands to work together.

Q4 What is a process in Linux?

Ans A process is a program that is currently running in the system. Each process has a unique process ID (PID).

Q5 What is the use of the ps command?

Ans The ps command displays information about currently running processes, including the PID, terminal & status.